

Gizzmo B1 Installation Manual



Safety warning: this device can cut and control engine boost. Incorrect settings can damage the engine. We recommend this be fitted by a professional technician and tuned on a dynamometer

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Welcome to the B1 Boost Controller

Thank you for choosing the Gizmo B1 — the latest evolution of a boost control platform refined over eight generations and thousands of units in the field. The B1 delivers closed-loop control across the full boost range, driving a genuine MAC 3-port solenoid at an adjustable frequency to hold your target precisely and repeatably. A 65,536-colour TFT LCD puts your engine vitals in front of you at a glance, while 2D mapping across up to 16 zones lets you offset boost against RPM, speed or voltage. Six independent memories can be switched on the fly to suit street, track or wet conditions, and active AFR protection with warnings and cuts is available via the optional module. Machined from billet alloy and designed, programmed and manufactured in Melbourne, Australia.



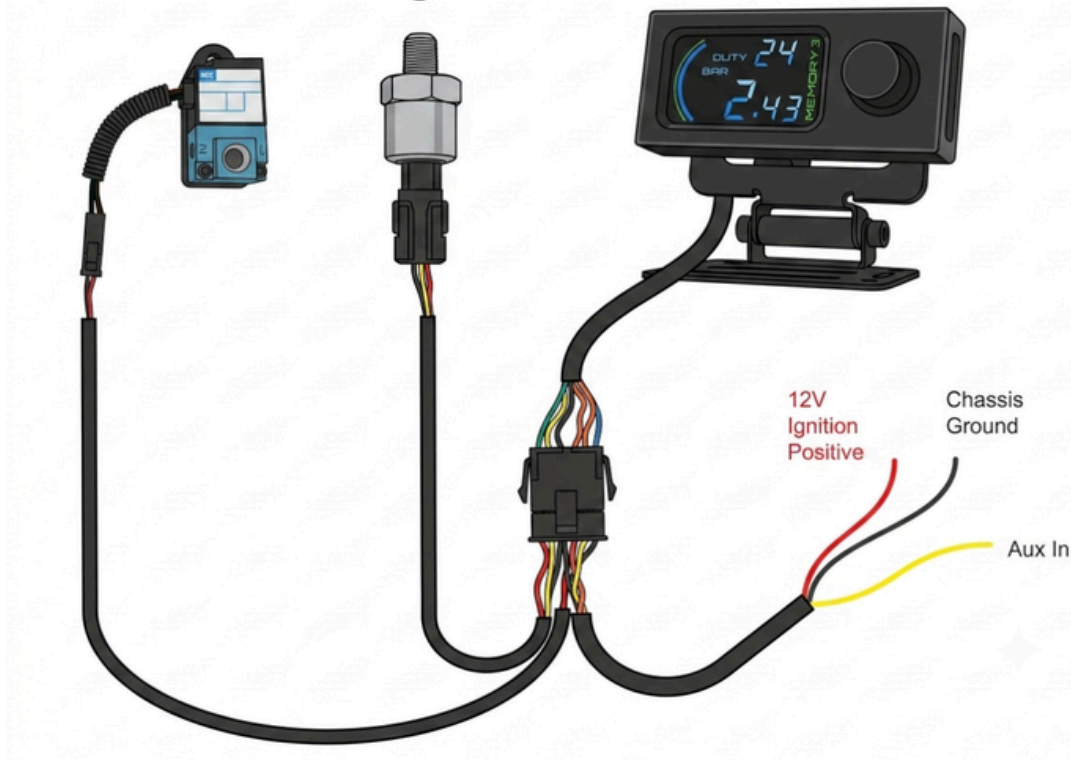
Technical Data



Function	Spec
Memories	6 with independent settings
Boost settings	Duty, Gate pressure, gain, antigain
Press Disply Values	PSI,Bar,KPA,
Vac Display Values	INHG,CMHG,KPA,PSI,Bar
Trims	RPM or Speed or Volts or % Volts
Max Pressure	3bar (up to 100psi with opt sensor)
Display	1.47 inch IPS LCD, 65K colour
Solenoid	MAC 3 port 5.4W
Case	CNC'd Billet Alloy
Origin	Australia

Installation

Gizzmo B1 Wiring

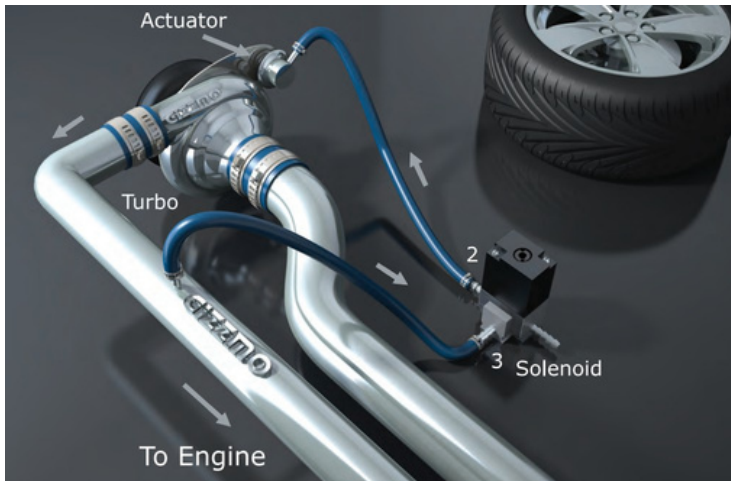


Please ensure the 12V ignition source is a fused supply, ideally off the fuse box or ignition switch. The Ground should be taken off of the chassis. The Aux in will depend on what option you're choosing to tie it to, however, this signal must **NOT** be connected to an ignition coil negative as this will destroy the B1. If connecting the B1 to an RPM source, it must be a low voltage digital signal e.g. ECU Tach output

Plan the harness route before you start pulling wires through. Wherever the loom passes through the firewall or any other panel, use an existing grommet if one is available, or fit a proper rubber grommet to a deburred hole – never pass a wire through a bare metal edge, as it will chafe through the insulation eventually and short out. Keep the loom well clear of exhaust manifolds, turbo housings, heater pipes and anything else that gets hot, and equally clear of moving parts such as the throttle linkage, steering shaft, pedals, bonnet hinges and cooling fans. Follow the factory harness where you can; it is already routed away from trouble and gives you plenty of tie-down points. Secure the loom every 200–300mm with cable ties or P-clips so it can't sag, swing or work its way into a hot or moving area over time, and leave a small service loop at the connector rather than pulling it tight. Keep the signal wires away from ignition leads, coils and other sources of electrical noise, and take your earth back to a clean, unpainted chassis point with a proper ring terminal. Finally, always disconnect the battery before making any connections, and check the full length of the run one last time before refitting the trim.

Wastegate Plumbing

Internal Wastegate Plumbing



External Wastegate Plumbing



Keep the runs as short and direct as practical, avoid low points where condensation or oil can collect, and route the hoses clear of exhaust components, turbo housings and moving parts. Assemble the solenoid as shown in the above picture with the supplied accessories. Mount the solenoid to a solid bracket with the ports facing downwards, away from direct heat, and allow enough hose length that engine movement doesn't tension the lines. Use the supplied silicone hose and hose clamps. Note the solenoid bracket is reversible should you want to bolt it from the bottom.

MAP Sensor Plumbing

Plumbing the MAP Sensor

The MAP sensor is what the B1 uses to measure actual boost, so its plumbing directly determines how accurately the controller can hold your target. Ideally, the sensor should be mounted inside the cabin, where ambient temperature is stable and free from the heat cycling and vibration of the engine bay — this gives the most consistent readings over a long drive or a track session.

Firstly, fit the supplied brass nipple adaptor to the MAP sensor using thread sealant or thread tape to seal the thread. Next, run the supplied silicone hose from the manifold through a suitable grommet in the firewall to the sensor, ensuring you have fitted the supplied filter inline and are using the supplied hose clamps, keeping the sensor itself secured to a solid mounting point rather than left hanging on the hose. Just as importantly, the sensor should have its own independent tap into the inlet manifold. Sharing a port with the blow-off valve, boost gauge, wastegate feed, or any other consumer introduces pressure fluctuations that the sensor will faithfully report, and the B1 will then chase those fluctuations as if they were real boost changes. A dedicated port costs a few minutes of work and removes the problem entirely.

If a dedicated port genuinely isn't available, tee the sensor in as close to the manifold as possible and as far as possible from any device that draws or vents air. Use good-quality reinforced hose rated for boost pressure, secure both ends, and keep the run as short and direct as practical — long runs add lag to the reading and dull the controller's response. Route the hose so it slopes continuously from the sensor to the manifold with no low points or sags where oil and condensation can pool, since a slug of liquid in the line will damp or block the pressure signal and contaminate the sensor. Keep the hose clear of exhaust components, turbo housings and anything that moves, and check the run for chafing wherever it passes through the firewall or over a bracket.

Bracket and Mounting the B1

Below shows how the bracket should be assembled



Choose a location where the display is easy to read at a glance and the rotary control falls comfortably to hand, without obstructing your view of the road, interfering with the steering wheel or controls, or sitting anywhere it could cause injury in an accident – check your local regulations on windscreen and dash-mounted devices before committing to a position. Avoid mounting the B1 in direct sunlight where possible: prolonged UV exposure and the heat that builds behind glass will shorten the life of any TFT display and can wash out the screen at exactly the moment you want to read it. A recessed spot low on the dash, under the overhang of the binnacle, or in a pillar pod all work well. Keep the controller away from heater and demister vents, and out of the engine bay entirely – it is designed for cabin use. Use the supplied brackets and mount to a solid panel rather than trim that flexes, so the unit stays put and the display doesn't blur over rough roads. Angle the screen towards the driver to minimise glare, allow enough slack in the harness that the connector isn't under tension, and route the wiring behind trim where it can't be snagged or chafed. Finally, plan the mounting position before you cut or drill anything – sit in the driver's seat, hold the unit where you intend to fit it, and check the view and reach in your normal driving position.

Navigation

There are two controls: the rotary knob and the push button (the knob is the button)

Action	Effect
Turn Knob	Move the blue selection dot / change the value being edited
Short press	Select the highlighted item, or start/stop editing a value
Long press (hold ~1.5 s)	Save everything and jump back to the home screen
Long press on the home screen	Open the Boost Setup screen

Reading the screen

- The blue dot marks the current line
- Grey text = not selected; green text = the active/chosen option; yellow = live data; light blue = screen title
- Dots appear along the bottom while a long press is being counted
- Turning the knob quickly steps values in larger increments (acceleration).
- Settings are written to memory when you long-press back to home

The Home Screen

Dual Display



Single Display



Single vs dual display layout (set in Display Properties)

What each area shows

- Main pressure reading, with unit label
- Pressure vs vacuum: the display switches automatically and can show different units for each
- Memory number (1–6), top right
- Side bar (configurable: boost, RPM or knock (with optional monitor))
- Second display field (dual layout only): solenoid duty %, max hold, or the aux input value

To change the memory: turn the knob while on the home screen.

Warning and cut indicators

- "w" in yellow = a warning threshold has been passed
-  Safety icon = a cut is active and the solenoid has been shut off

Before you start

It is important to set the boost cut prior to trying to setup the boost, not only for your engines safety, but because of the structure and dependances of the firmware. For the best results, we suggest setting it not to far above the boost set point e.g. ~4psi over the set point.

From the home screen, pushing the knob once will take you to the 'Main Menu'

Main
menu



While you're here, you may want to enter 'Display' to select your preferred units. Simply scroll you whichever pressure and vac you want and select each one. After this you'll have to exit (push and hold) to return to the Home screen



From home, push select to return to the 'Main menu'. Scroll to and select 'Engine Safety'. Select 'Warning' and adjust to 1 or 2 psi below where you intend to set the boost cut. Unselect 'Warning'; scroll, select and then adjust 'Boost cut' in the same manner and then push and hold to escape back to the Home screen

Setting Boost #1

Boost Setup
menu



In the 'Home Screen', turn the knob to select the memory number you wish to set or adjust. Push and hold the knob until the 'Set Memory' screen appears as above. If you wish to adjust the boost, push the knob with 'Duty Setup' highlighted to enter the boost setting screen as below.

Duty directly corresponds to boost e.g. more duty = more boost

Duty Setup
logger



If you are 100% confident of the boost to duty relationship, you can select 'User Set' and adjust it to the known boost setting and then unselect it, turn the knob to highlight DC (Duty Cycle), select and adjust to the known duty, then unselect it and move to 'Set' and select it to set the memory setting. If the relationship is unknown, turn the knob to highlight DC and start at a low duty (~10 to 15%). At this point you need to apply a load on the engine and will see a screen like the one below.

Duty Setup
logger



Setting Boost #2

With the engine under load with the above screen, you can turn the knob to adjust the dutycycle ('DC') in real time and see the response via the 'BST' level. It is **important** that you have had the boost stable on the dyno like line on the screen so that the B1 can calculate the 'stable' boost level for the set duty. After you release the load off the engine, you will be graced with the screen below.

Duty Setup
logger



If you achieved a stable boost in the previous procedure, the 'LEVEL' and 'USER SET' on the display will show the stable boost level for the duty set. If you for any reason disagree with the value, or the boost was not stable for long enough for the B1 to capture it, you can select 'User Set' and adjust it inline with the level you observed.

On completion of setting the boost and duty, you must either Set or Cancel the settings to escape to the 'Set Memory' screen.

Now, to explain the remaining boost settings.

GATE PRES: This is expressed in boost pressure. When the vehicle goes above Ovac, the B1 will hold the wastegate closed e.g. attempt to reduce lag as much as possible until the gate pres setting is reached, at which point it will switch to the gain setting (explained shortly). If your boost is set, for example, at 15psi, a typical gate setting to start with would be ~7psi. This should be set after duty.

GAIN: This is expressed as a percentage where 0% is minimum and 100% is maximum (more aggressive). The gains function is to reduce turbo lag. If this setting is too high, your boost may oscillate, be unsteady and/or spike when coming on boost. This should be set after gate pres. Increase this slowly e.g. increments of 5%, until your boost oscillates, gets unsteady or spikes when rolling on.

ANTIGAIN: This is like gain, but for when boost goes over the set pressure. Essentially, this is how quickly duty is decreased if boost goes over the set pressure. 0% means it will not decrease the duty over the boost pressure, whereas 100% means it will decrease duty a lot if it goes over the set pressure. A typical setting would be 10%; however, if your vehicle has boost creep or other mechanical issues, a higher setting may be in order.

If you adjust the duty cycle/boost, it may be best to revisit the 3 setting above to optimise the new setup. Also note these setting are independent to each memory setting e.g. it is NOT the same for all memories.

Setting Boost #3

Boost Analyser
Setup menu



In the previous boost setup screen, select and enter 'Options...'; the screen should look like the one above. Set this according to your preference. The features are...

RETURN: Takes you back to the boost setup screen

LOG TYPE: Clicking this will swap between SINGLE and CONT. Soon we will go over the boost analyser feature which works in a similar way to a dyno. The difference between SINGLE and CONT is that after you have logged a boost run, in SINGLE you cannot clear and start another run until you have clicked the knob whereas in CONT, every time you take you re-enter boost, it will resart a new run. It is a lot easier to understand this once you have seen how the boost analyser works.

MAX LOG TIME: On the Boost analyser screen below, the horizontal time axis is set by this value, which in the screen below is set to 4sec. So if, after reaching $\frac{1}{2}$ of your gate pressure, your engine takes 6 seconds to reach your shift point/redline, you would set this to 6sec. You dont have to complete the run, if it sees you abort at any point, it displays the log up to that point.

BOOST TYPE: This selects between 1 DIM (1 dimensional boost e.g. normal) and 2 DIM which is where you can offset the boost based on your aux input. IMPORTANT: 2-dimensional boost is only available when the aux input is set to

- Volts raw or percentage
- RPM
- Speed

Boost Analyser



Boost Analyser



The purpose of the Boost Analyser is to assist you in setting up your gain, antigain, gate and 2-dimensional boost, if being used.

The Boost Analyser automatically starts when...

1. You are in the Boost analyser setup menu or Boost setup menu or 2-Dimensional mapping screen
2. You apply more boost than the gate pressure setting divided by 2

The Boost Analyser stops when...

1. You remove boost going ~0vac or less or
2. You reach the end of the x axis time scale (capped at 4sec above)

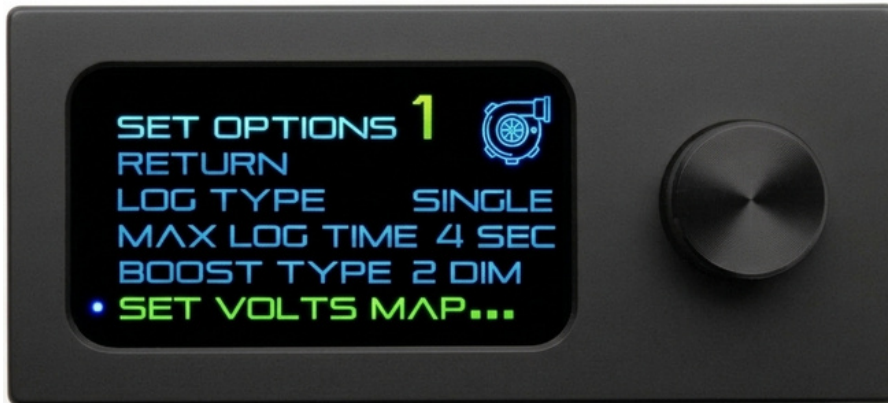
Once the B1 has logged a boost run, you may turn the knob to scroll a vertical timeline. All the values at the top of the screen will show you their values at the point of the time line. Please note that the TRIM on the top display will only show if you're running 2-dimensional boost.

On the vertical axis, the yellow digits indicate pressure and the white digits, when present, indicate the aux in value.

you can exit the Boost analyser screen at any time by just clicking the knob.

2-Dimensional Boost

Boost Analyser
Setup menu



If you select 2 DIM in the boost analyser setup menu, a new line will appear which will say 'SET X MAP...' where X will be RPM, Volts or Speed. If you then select this line, it will take you to the 2-Dimensional mapping screen below. In this example I will use volts, but all the same principals apply to speed and RPM.

Trim Setup
Page



In this screen, turning the B1 knob cycles from 'RETURN' (which returns you to the Boost analyser Setup menu), through all the adjustable zones, to setup which we'll cover shortly.

When you select an adjustable zone, the white box highlights the setting and the zone is displayed on the 1st line below settings. In this example is shows as VOLT ZONE 0.0V being the 0.0V zone, which presently, as per the example, is set to offset the duty by 0%. If the text is in red, like the 5 next to the 0 setting, this means it will pull 5% OFF the duty (reduce the boost), whereas if the text is in green, it will add the value to the duty (increase the boost). 0 is always displayed in white.

If you click the knob when you have an adjustable zone highlighted, the blue ball associated with that zone will fade, the VOLT ZONE X.X will also fade, and the ADJ X% will highlight to show you are now in an adjustment mode. If you now turn the knob, the value(offset) will adjust the duty associated with that zone. This will be displayed in 3 ways; the 1 row table value, the ADJ X% value, and the ball height. If you are not at the original setting of when you entered that zone e.g. you have adjusted it, the ball will be yellow. If you take it back to the original setting, the ball will change back to faded blue. Please note that the B1 interpolates between zones for a smooth transition.

2-Dimensional Setup



This menu is pretty straightforward and self-explanatory. Again, the option settings are exactly the same for RPM and Speed simply being, for example, MIN SPEED or MAX RPM etc
The four options are ...

MIN VOLT: This will be the minimum zone on the Trim Setup Page (shown on page 14)

MAX VOLT: This will be the maximum zone on the Trim Setup Page (shown on page 14). It's important to note that all the zones between the MIN and MAX will be evenly divided e.g. if MIN RPM was 4000rpm and MAX RPM was 8000rpm and ZONES (below) = 9, the zones would be spaced 500rpm apart.

ZONES: The number of zones in total spaced evenly between the MIN and MAX settings. The B1 interpolates between zones, so even if you are not running many zones, it won't suddenly jump between them. The minimum number of zones is 2 and maximum is 16.

RETURN: This will return you to the Trim Setup Page

Display Setup

Display Menu
#1



From the home screen, click the knob once to go into the main menu (page 11) and then press it again to enter the display menu above. Once here, turning the knob will cycle through the pressure options, then NEXT option, then the vacuum options. Clicking the knob when on a pressure or vac option will switch the realtime pressure/vac display to that measurement type. If you click 'NEXT' you will get the menu below.

Display Menu
#2



The following options are...

BACKLIGHT: Adjusts screen brightness in 1% increments up to 99%

DISPLAY: This option toggles between single and dual displays as shown on page 10

VAC DISP: When this is ON, the B1 will display vacuum as well as boost. When OFF, it will only display Ovac upwards.

SIDE BAR: For the sidebar on the left side of the screen, it by default will display boost. There are 2 other option available, but there are requirements. If you have the aux in set to RPM, the menu option will let you select to display RPM with the max segment inline with your shift RPM. The last option is if you have a knock monitor with a 0 to 5v output fed into the aux in, in which case it will display engine noise.

TOP DISPLAY: Clicking the knob cycles through the 3 options for the top display which are SOL % for real time solenoid duty cycle, MAX which holds the max boost for a second after backing off and AUXIN which displays your auxillary in value.

Engine Safety #1



Engine safety for any metric is treated in 2 steps being warning and cut. Warning will be displayed, where possible by turning the respective digits green in colour. Where not possible e.g. an aux in warning on single display mode, will display a W to the right of the sidebar. In any case. A cut will set the duty cycle to 0% to dump all boost and will display the yellow safety symbol as above.

The boost warning and boost cut are always available, but the auxiliary input warnings are only available when the aux in is employed (except the offset and speed aux in options), and the options vary based on which aux in is employed. If no aux in is enabled, AUX OPT and AUX SAFETY above will not be visible. Also, if AUX OPT is set to OFF, AUX SAFETY will not be displayed. However, if AUX OPT is ON and you click on AUX SAFETY, depending what aux in is selected, there are several scenarios. For RPM, there is just one setting for RPM CUT. For knock there are 2 settings, Knock warning and Knock cut, both set by %. Volts and PWM are always set in the same manner, for this example I will use Duty...



The voltage raw & percentage, and pwm ms & duty, the above screen share all the same options.

WARN IF: This toggles between ABOVE and BELOW. In the example above the warning will display if the aux in duty cycle goes above 89%. The following line (89%) can then be adjusted to suit your needs.

CUT IF: Exactly as above, except it will display the safety icon and cut the boost as well.

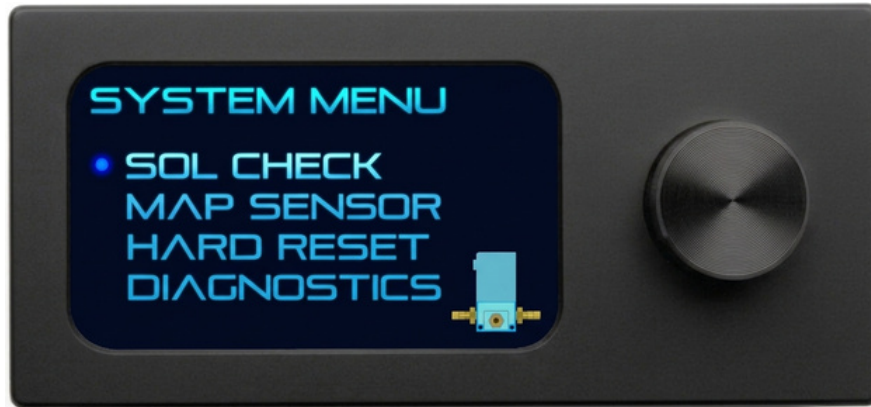
Engine Safety #2



The first couple of options are obviously to set your warning and cut thresholds, however, the third option 'APPLY WITHIN XX% OF SET BOOST' is important. What it is representing is the percentage below the set boost in which this warning and cut will be active. For example, if my boost memory is set to 20psi for the above screen. 10% of 20psi 2psi; so if I go leaner than 12.2afr WHEN I am over 18psi (20psi - 10% (2psi) = 18psi), I will get a warning and at 12.5afr is still over 18psi, it will drop the boost.

System Menu

System Menu
Screen



You get to the System, you go through the Main Menu (page 11). As above, there are 4 options.

SOL CHECK: When the B1 powers up, there is the option to pulse the solenoid a few times as a way of confirming everything is OK. If you click on the SOL CHECK option you can adjust the number of clicks or even set it to 0 to turn it off.

MAP SENSOR: Clicking on this option takes you to the menu where you can either select an alternative MAP sensor or setup a custom one. This is covered in more detail on the following page.

HARD RESET: Entering this menu will show 3 options for what to reset. These are Memories, Settings or Both. After selecting one, there will be a confirmation window to confirm before resetting.

DIAGNOSTICS: This enters a screen to check some basic hardware inputs into the B1 including voltages, button, knob and FW version date as below.



MAP SELECTOR



The top 4 options are presets, and the 5th is to configure your own sensor.

GIZZMO 3BAR is the sensor that your B1 was supplied with

BOSCH 119 OR 076 are the bosch O 261 230 119 and O 281 006 076 which both have the same calibration.

SSI XXPSI 4.5V are SSI sensors but please note, these are 0.5v to 4.5v, NOT 1v to 5v

CUSTOM will take you to the screen below to configure any MAP sensor with a range inside of 0v to 5.0v



MAP sensor documentation always has a low point and a high point. An example of this is a 3bar pressure sensor with a span of 0.5V to 4.5V. This means that 0bar = 0.5V and 3bar = 4.5V which is how you go about setting this screen

BAR PSI KPA: Clicking the knob cycles through these 3 options so you can select the metric that your MAP sensor's data sheet is calibrated in.

L VOLTS: Set this to the low voltage calibration point

L PRESS: Set this to the low pressure calibration point

H VOLTS: Set this to the high voltage calibration point

H PRESS: Set this to the high pressure calibration point

Auxillary Input Setup



Above is your Auxiliary Input screen (accessed via HOME > MAIN MENU > AUX IN). Regardless of your selection, the window below will ask you to confirm your choice. If you select 'NO. BROWSE ONLY', you can go and have a play with the settings within your selected option without actually changing any of your present settings. This gives you an excellent opportunity to get a feel for an option before committing. If you select KNOCK and confirm it, there will be no options as knock is preset to 0% = 0V and 100% = 5.0V

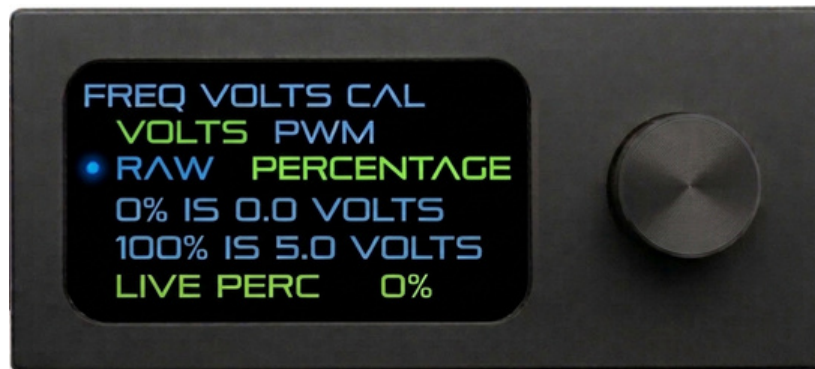


If you select yes, everything you do will be applied immediately. Because of the options available, we will outline them below.

Volts Input Setup

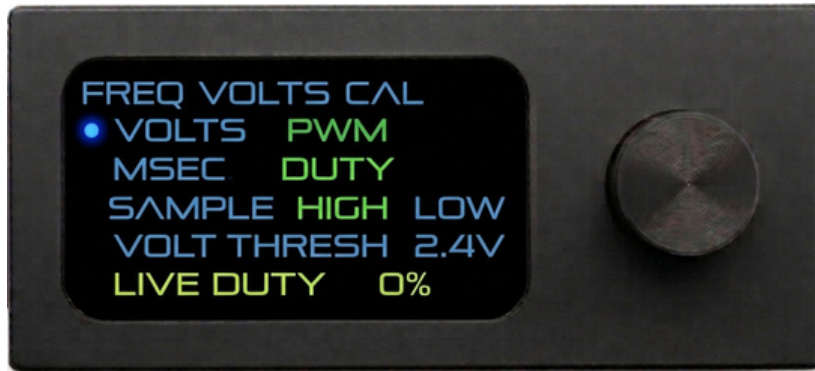


Above are your two voltage options, RAW and PERCENTAGE. Raw just means display as a voltage without any alteration e.g. 1.2V on the aux in will display as 1.2V. The LIVE display at the bottom of all option screens is what will be displayed on the top display based on the setting as they are right now.



If you select percentage, for something like a position sensor, you will have two settings to configure, 0% and 100%. This aligns a voltage to 0% and 100%. An example of this would be if you set 0.5V to 0%, and 4.5V to 100%, then anything on your aux in wire below 0.5V will display as 0% and 4.5V will display as 100% so 2.5V (half way between 0.5V and 4.5V) will display as 50%.

PWM Input Setup



PWM, like voltage also comes in two flavours, MSEC (milliseconds) and DUTY. As can be seen from the screens above, they have the same settings.

SAMPLE: This has two options, HIGH and LOW. If HIGH, the milliseconds/duty will measure the part of the aux in signal. so the LOW setting measures the part of the aux in signal.

VOLT THRESH: Set this voltage to the transition point you want to establish to define LOW vs HIGH. An example using the above images is anything above 2.4V is defined as high and anything below 2.4V is considered low.

RPM/SPEED Input Setup



RPM and Speed are set in a similar manner.

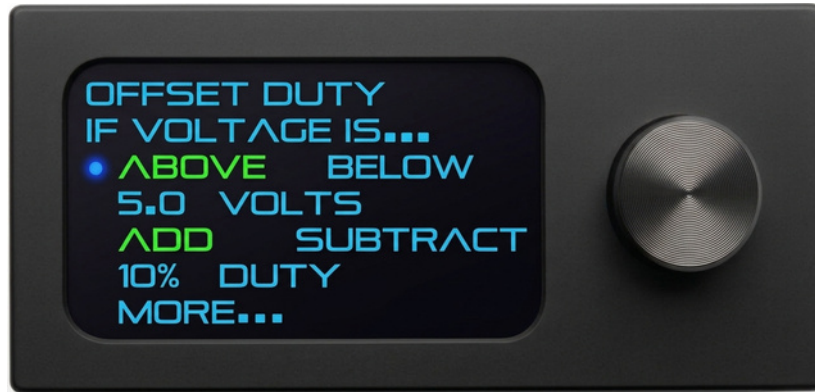
COMMON: RPM and, less so, speed have common relationships between frequency and RPM/speed. For example, any two six-cylinder engines with a single coil will have the same tach frequency. The LIVE data on the bottom line will give you real-time updates of the displayed RPM/Speed based on your adjustments. If the common frequencies do not give you the correct RPM/Speed, scroll to the COMMON CUSTOM line and click to toggle to CUSTOM. From here, scroll to the frequency line below COMMON CUSTOM and click the knob once. This allows you to adjust the frequency for the RPM displayed to the right of the frequency. If you click the knob again, you can adjust the RPM for the frequency on the left. Clicking the knob again deselects both. So, as an example, if you know that the tach frequency is 45hz at 2200rpm, you would select CUSTOM, scroll to the frequency and click once, turn the knob to adjust the hz to 45hz, click again to select RPM and turn the knob to set to 2200RPM.

SHIFT RPM: This sets the shift RPM. The shift RPM changes the RPM to yellow in the same way as the safety warning works.



The only additional thing to mention here is that clicking on the KPH MPH line toggle between the two. This will also display the selected option on the top display of the home screen and any other speed related options.

Offset Duty (scramble) Setup #1



Offset can do scramble, but it way more capable with a ton more flexibility. One very **IMPORTANT** caviat is that if you are using the BELOW option, ensure that whatever is connected to the aux in has a positive signal on it when not triggered.

ABOVE BELOW: clicking toggles this between the two options. ABOVE will make the offset trigger whenever it goes 'above' the voltage set below this option; in the case of the image above, 5.0V. So for the example image above, whenever the aux in wire goes above 5.0v, the offset will fire. BELOW does the opposite and triggers whenever the voltage goes 'below' the set voltage.

VOLTS: This sets the threshold voltage at which to trigger the offset (explained above). It spans from 0.5V to 15.0V

ADD SUBTRACT: Clicking toggles this. ADD means that it will 'add' the duty setting below this option on top of the present duty cycle when the offset is triggered, whereas SUBTRACT will remove duty (lower boost) when the offset is triggered.

DUTY: The amount of duty the offset function will add or subtract from the present duty cycle when triggered

MORE...: This take you to the next screen outlined on the following page

Offset Duty (scramble) Setup #2



There are two options for how we handle the operation of offset when it has been triggered,, the first one we'll cover is 1 SHOT

TRIG MODE: Trig mode has two options, 1 SHOT and LATCHED, we'll cover latched later on.

DUTY CONT: Duty Cont can toggle between two functions, FIXED and DECAY. FIXED means it will hold whatever duty cycle offset you've set for the TIME ON period and then remove the offset, whereas DECAY will initially add/subtract the offset and then slowly release it over the set TIME ON period. An example of this would be if you had your boost set to 14psi which aligned with a 20% duty cycle, and your offset was set to add 5% with 1 shot and decay over 10 seconds. When triggered, your boost would go from 20% to 25% and then drop 1% every 0.5 seconds until it again reached 20%

TIME ON: This is how many seconds the offset will be active for once triggered



Latched mode has no time limit. As an example, if offset is set to trigger if above 5.0V and remove 10% duty, then whenever the aux in wire is under 5.0V, 10% will be remove indefinitely until it goes below 5.0V

Solenoid Frequency



Duty cycle sets the proportion of each cycle the solenoid is energised, and is varied automatically by the boost control loop. Frequency sets how often that cycle repeats. It is a fixed setting: whatever value you enter is used at all times and under all conditions, and the controller never alters it. Setting frequency does not raise or lower boost by itself – it changes whether a given duty is delivered as a few large slugs of air per second or many small ones. The B1 accepts any value from 10 to 50 Hz. Low frequency (10–20 Hz). The valve completes its full stroke every cycle, giving full flow authority and predictable behaviour at the extremes of the duty range. It also tolerates a slow, worn or sticky solenoid. The cost is pulsation: visible actuator flutter, boost hunting around target, coarse steps between duty settings, a noisy MAP signal that makes the loop harder to tune, and audible buzzing from the solenoid.

High frequency (35–50 Hz). The valve hovers part-open rather than fully stroking, throttling the bleed instead of pulsing it. Control is smoother and quieter, with finer resolution and a cleaner feedback signal. The cost is reduced flow – you may not reach target boost even at 100% duty – plus a dead band at low duty, saturation at high duty, and greater sensitivity to supply voltage and coil temperature, so the duty required for a given boost can drift as the car heats up.

Recommended starting point: 30 Hz for the MAC 3-port solenoid. Tune the control loop at this setting first. Raise in 5 Hz steps if you see actuator flutter, coarse boost steps or solenoid noise; lower in 5 Hz steps if you run out of duty authority, find a dead band at the ends of the range, or see the required duty drift with temperature.

Changing frequency changes the duty needed for a given boost level. After any change, re-verify your base duty and closed-loop calibration at low boost before working up.